

Joshua O'Callaghan

+1 (323) 393-3483 • jdocalla@asu.edu • www.jocal.dev • linkedin.com/in/joshuaocallaghan

EDUCATION

M.C.S. Computer Science (Big Data)

Dec 2020

Arizona State University, Ira Fulton School of Engineering, Tempe, AZ

B.S. Computer Science (Cyber Security)

Dec 2019

Arizona State University, Ira Fulton School of Engineering, Tempe, AZ

Cumulative GPA: 3.61 | Major GPA: 3.78

Relevant Coursework:

Data Structures and Algorithms, Algorithm Design & Analysis, Data Mining, Distributed Databases, DBMS Implementation

PROFESSIONAL EXPERIENCE

Intel – Software Engineering Intern

Sept 2019 – Present

- Building factory reports using Power BI and SQL
- Assembling various scripts and processes to help automate daily tasks using Python
- Visualizing data from factory tools to ensure quality and identify errors with Plotly

American Express – Software Engineering Intern

June 2019 – August 2019

- Redesigned infrastructure for Platform Engineering team
- Introduced new tools such as Kubernetes to improve operations
- Administered 4 server nodes running unique applications & tools on a HA Kubernetes cluster
- Created automation scripts in Bash to take care of simple tasks that were previously manual
- Recorded the install process of Kubernetes and related tools for team archives

Arizona State University – UGTA (Principles of Programming Languages)

Jan 2019 – May 2019

- Mentored students in advanced conceptual topics and critiqued C++ code

SKILLS

Coding Languages: Python, Java, C++, C, C#, HTML, CSS, JS, Bash, R, Scala, Scheme, Prolog

Software: Visual Studio, JetBrains Products, Azure

Operating Systems: Windows, Linux (Ubuntu, Mint, Cent OS, Red Hat)

Command Line Tools: GDB, Netstat, Ipconfig

Big Data: Kubernetes, Docker, Elasticsearch, MapR

Soft Skills: Leader, Communicative, Self-motivated, Problem Solver, Innovator

Other: Git, Scrum/Agile Development, Amazon Web Services, Gephi, Opencv, Docker, Pandas, Plotly, Python-pptx, D3js, PostgreSQL

ACADEMIC PROJECTS

Distributed Databases | Individual Project: Distributed Database Techniques

Spring 2019

- Constructed a DB with Python functions to partition using various horizontal fragmentation approaches
- Generated functions to insert, query, sort and join the data from all partitions

DBMS Implementation | Team Project: BigTable Implementation

Spring 2019

- Collaborated in a team of 6 to convert an existing relational DBMS into a Bigtable-like DBMS
- Refactored existing functions and wrote searching/ sorting functions to maintain DB throughout insertion

Capstone Project | Team Project: General Dynamics HMI Capstone

Fall 2019

- Operated in a team of 5 using Agile Development with General Dynamics
- Analyzed Linux HMI frameworks to fabricate a GUI for a classified communication device

Capstone Project | Team Project: Construction Simulator in AR

Spring 2019

- Developed in C# using Unity and Microsoft HoloLens to help students simulate construction projects

PERSONAL PROJECTS

Online Portfolio | Personal Website

Fall 2019

- Designed a website using HTML5, CSS, and JS
- Managing an AWS EC2 instance running Apache on Ubuntu 18.0.4 to host domain

Face Recognition | OpenCV Python Project

Fall 2019

- Explored ML Library, OpenCV, to recognize a human face with a computer webcam

EXTRACURRICULAR

Google HashCode Competition

Feb 2020

- Placed 44th in US, 844th globally as a team of four in Google HashCode programming competition

National Society of Black Engineers | Corporate Liaison

Fall 2018 – Present

- Arranged sponsorships with large companies and organized meetings for the general body